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Assignment #2

Mathematics - III (MTH 231)

Date: September 05, 2024

Branch - Electrical Engineering

Section - I and II

1. Given the points $(1, 2)$, $(3.5, 7)$, $(-43.8, -87.6)$, $(0.392, 0.784)$, $(6, 12)$ and $(0, 0)$ in the plane. The polynomial $y = 2x$ passes through these points. State which of the following statements are true or false and support your answer with reason.
 - (a) I can find a polynomial of degree 5 passing through the above points.
 - (b) I can find a polynomial of degree 6 passing through these points.
2. Find a polynomial passing through the following points $(-2, -6)$, $(-1, 0)$, $(1, 0)$, $(2, 6)$, $(4, 60)$.
3. Find a polynomial that intersects the curve $\sin\left(\cos(e^{x^2})\right)$ at $x = 0$, $x = -1.5$, $x = 3.7$, $x = 8.6$, $x = 20.9$ and $x = -46.123$.
4. A polynomial $p(x)$ intersects the parabola $y = x^2$ at 7 points. What can you say about the degree of $p(x)$? Justify your answer.
Hint : If two polynomials $p(x)$ and $q(x)$ intersect each other, the intersection points are the roots of the polynomial $(p - q)(x)$.
5. Find the polynomial defined as follows

$$p(n) = 1^5 + 2^5 + 3^5 + \dots + n^5$$

where $n = 1, 2, 3, \dots$ run over the set of natural numbers.

Hint : If you interpolate a polynomial of degree 7 using 8 or more data points, you get the polynomial back as the answer. Here $p(n)$ will be a 7 degree polynomial.

6. The melting point of an alloy of Lead and Zinc was experimentally measured and the following observations were made

Percentage of Lead in alloy	Melting Point of the alloy ($^{\circ}\text{C}$)
40	180
50	204
60	226
70	250
80	276
90	304

- (a) Find the melting point of the alloy containing 16% Zinc.
- (b) What percentage of Lead needs to be mixed in so the alloy has a melting point of 260°C .

7. The following table gives the value of the integral, where $g(\Phi)$ is a function of Φ

$$F(\Phi) = \int_0^{\Phi} \frac{d\Phi}{g(\Phi)}$$

Φ	21°	23°	25°	27°	29°
$F(\Phi)$	0.3706	0.4068	0.4433	0.4216	0.3982

- (a) Find Φ such that $F(\Phi) = 0.4301$.
 (b) Approximate $F(\Phi)$ for $\Phi = 35^\circ$.
 (c) Approximate $F(\Phi)$ for $\Phi = 20^\circ$.

8. Apply appropriate interpolation formula to find the sale for the year 1966 given that

Year	1931	1941	1951	1961	1971	1981
Sales (in Lakhs)	12.48	15.85	20.13	27.82	39.17	52.25

9. Given the integral $f(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-x^2} dx$ and its values for different values of x . Find the value at $x = 0.6538$ and 0.6483 using the appropriate central interpolation formulae.

x	0.62	0.63	0.64	0.65	0.66	0.67	0.68
$f(x)$	0.6194114	0.6270463	0.6348570	0.6420292	0.6493765	0.6566275	0.6637820

10. Find the value at $x = 656$ with the following data points using

- (a) Lagrange's Formula
 (b) Laplace-Everett Formula

x	654	658	659	661	672	681	683
$f(x)$	2.8156	2.8182	2.8189	2.8202	2.8274	2.8331	2.8344

11. Approximate the value of $\sin(x^2)$ at $x = 0.35$ using the following data. Compare your value with the actual value and find the relative error.

x	0.2	0.5	0.8	0.11	0.3	0.15	0.78
$\sin(x^2)$	0.000628	0.003926	0.01005	0.00019	0.00141	0.000353	0.009556

12. Determine the missing values in the divided difference table provided below. Also find the corresponding polynomial.

x	$f(x)$	1 st Order difference	2 nd Order difference	3 rd Order difference
0	1	2 y 0	x 0	z
1	3			
2	3			
3	w			

13. Let α be a real number. If the fourth order divided difference of the function $f(x) = \alpha x^4 + 5x^3 + 3x + 2$ at the points $x_0 = 0.1$, $x_1 = 0.2$, $x_2 = 0.3$, $x_3 = 0.4$ and $x_4 = 0.5$ is 5. Then find the value of α .
14. An ISRO - DRDO joint collaboration conducted a test of an advanced supersonic missile code named Project 4A0S. Through satellite tracking as well as onboard telemetry instruments, the team tracked the speed and height of missile at regular intervals and obtained the following data

Time after launch (in sec)	Speed (in km/sec)
3000	0.2548
4000	0.6878
5000	1.6378
6000	2.6256
7000	2.7431
8000	2.7625
9000	2.5900

- (a) Calculate the speed of the missile at $t = 6250$ seconds after launch using suitable method.
- (b) A primary aim of the mission was to establish when the missile broke Mach8 speed barrier (around 2.72km/sec). Calculate the time at which the speed of the missile became 2.72km/sec.
15. Given that a function f satisfies the following conditions
- (a) $f(0) = 1$, $f(1) + f(2) = 10$ and $f(3) + f(4) + f(5) = 65$.
- (b) All second order differences are equal.

Find the value of the function at $x = 4$.